

[Aim: 100/100 in Maths]

आरंभ

# POLYNOMIALS

**Lecture #2**

# ATK<sup>2</sup>

→ A · E

→ [variable]<sup>whole no.</sup> → Polynomial.

→ n → 1 → monomial  
→ 2 → binomial  
→ 3 → trinomial

→ D → 1 → linear  
→ 2 → Quadratic U ∩  
→ 3 → cubic

P(x) = x<sup>2</sup> - 2  
↑  
a polynomial in variable 'x'.

अगर further solvable है  $\rightarrow$  तो complete solve के लिए decision ले

# LP: Which of the following is/are polynomial?

(i)  $\frac{x^3 - x}{\sqrt{x}} \Rightarrow \frac{x^3 - x}{x^{\frac{1}{2}}} \Rightarrow \frac{x^{\textcircled{3}}}{x^{\textcircled{\frac{1}{2}}}} - \frac{x^1}{x^{\frac{1}{2}}} \Rightarrow x^{3-\frac{1}{2}} - x^{1-\frac{1}{2}} \Rightarrow \boxed{x^{\frac{5}{2}} - x^{\frac{1}{2}}}$  No

(ii)  $\underbrace{(x)^{1/2}}_a + \underbrace{(x)}_b \underbrace{(x)^{1/2}}_a - \underbrace{(x)}_b \Rightarrow (x^{\frac{1}{2}})^2 - (x)^2 \Rightarrow x^{\frac{1}{2} \times \frac{1}{2}} - x^2 \Rightarrow \boxed{x^1 - x^2}$  Yes ✓

(iii)  $\frac{\sqrt{x^4 + x^3}}{x} \Rightarrow \frac{(x^4)^{\frac{1}{2}} + x^3}{x} \Rightarrow \frac{x^{4 \times \frac{1}{2}} + x^3}{x} \Rightarrow \frac{x^2 + x^3}{x} \Rightarrow \frac{x^2}{x} + \frac{x^3}{x} \Rightarrow x^1 + x^2$

(iv)  $1/x \Rightarrow \frac{1}{x} \Rightarrow x^{\textcircled{-1}}$  No

$\Rightarrow \boxed{x^1 + x^2}$  Yes ✓

#LP: For what values of  $n$  is the expression a polynomial?

$$p(x) = \frac{x^n + x^2}{x}$$

$$p(x) \Rightarrow \frac{x^n}{x^1} + \frac{x^2}{x}$$

$$p(x) \Rightarrow x^{n-1} + x$$

$\in$  belongs to  
 $\notin$  does not belong to

$$n=0 \rightarrow x^{0-1} + x \Rightarrow x^{-1} + x$$

poly x

$$n=1 \rightarrow x^{1-1} + x \Rightarrow x^0 + x = 1 + x \checkmark$$

$$n=2 \rightarrow x^{2-1} + x \Rightarrow x^1 + x \checkmark$$

$$n=3 \rightarrow x^{3-1} + x \Rightarrow x^2 + x \checkmark$$

$n \rightarrow$  all natural Number

$$N = \{1, 2, 3, \dots\}$$
$$\boxed{n \in N}$$



# # Value of Polynomial:

$$P(x) = x + 1$$

Find the value of poly. at  $x = 4$ ?

ans  $P(4) = 4 + 1$

$$P(4) = 5$$

(i)  $f(x) = x^2 - 1$ , at  $x = -1$

$$f(-1) = (-1)^2 - 1$$

$$\Rightarrow 1 - 1$$

$$f(-1) \Rightarrow 0$$

$$P(-1) = -1 + 1 \Rightarrow 0$$

$$P(0) = 0 + 1 \Rightarrow 1$$

$$P(t) = t + 1$$

$$P(\text{shobby}) = \text{shobby} + 1$$

(ii)  $f(x) = 2x + 1$ , at  $x = \frac{1}{2}$

$$f\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) + 1$$

$$= 1 + 1 \Rightarrow 2$$

# #~~4~~ Zeroes of Polynomial: Zeroes/Root/Solution

→  $x$  की वो value जिससे  $P(x)=0$  हो जाए।  
 $y=0$

#LP:- Find the zero of polynomial

$$P(x) = 3x - 3$$

$$x=1 \rightarrow P(1) = 3(1) - 3 = \underline{\underline{0}}$$

zero of  $P(x)$ .

#LP:-  $g(x) = 3x - \frac{9}{2}$

$$\frac{g(x)=0}{3x - \frac{9}{2} = 0}$$

$$3x = \frac{9}{2}$$

$$x = \frac{3}{2}$$

zero of  $g(x)$

# Q:-  $f(x) = \frac{14}{15}x - \frac{1}{4}$

$$\frac{f(x) = 0}{\frac{14}{15}x - \frac{1}{4} = 0}$$

$$\frac{14}{15}x = \frac{1}{4}$$

$$x = \frac{15}{56}$$

(अथ)

जिस भी poly. को zero निकालना है उसे  $poly = 0$  कर दो।  
फिर जो  $x$  की value आएगी वह zero होगा।

→ zero of  $f(x)$

#LP: If one of the zeroes of polynomial  $\underline{a^2x^2 + x + b^2}$  is -1 then :

(A)  $a^2 + b^2 = 0$

(B)  $a^2 + b^2 - 1 = 0$

(C)  $a^2 - b^2 + 1 = 0$

(D)  $a^2 + b^2 = -1$

let  $P(x) = a^2x^2 + x + b^2$

$$P(-1) = 0$$

$$a^2(-1)^2 + (-1) + b^2 = 0$$

$$a^2 - 1 + b^2 = 0$$

$$a^2 + b^2 - 1 = 0$$



# LP: If 2 and 0 are the zeroes of the polynomial  $f(x) = 2x^3 - 5x^2 + ax + b$  then find the values of a and b.

$$f(x) = 2x^3 - 5x^2 + ax + b$$

$$f(2) = 0$$

$$2(2)^3 - 5(2)^2 + a(2) + b = 0$$

$$16 - 20 + 2a + b = 0$$

$$-4 + 2a + b = 0$$

$$2a + b = 4 \quad \text{--- (I)}$$

$$2a + 0 = 4$$

$$a = 2$$

$$\Rightarrow a = 2 \quad \checkmark$$

$$f(0) = 0$$

$$2(0)^3 - 5(0)^2 + a(0) + b = 0$$

$$0 - 0 + 0 + b = 0$$

$$b = 0 \quad \checkmark$$

# # Geometric Meaning of Zereeos of Polynomial:

# LP: Draw the graph of  $P(x) = 2x - 4$  and identify the zeroes of this polynomial.

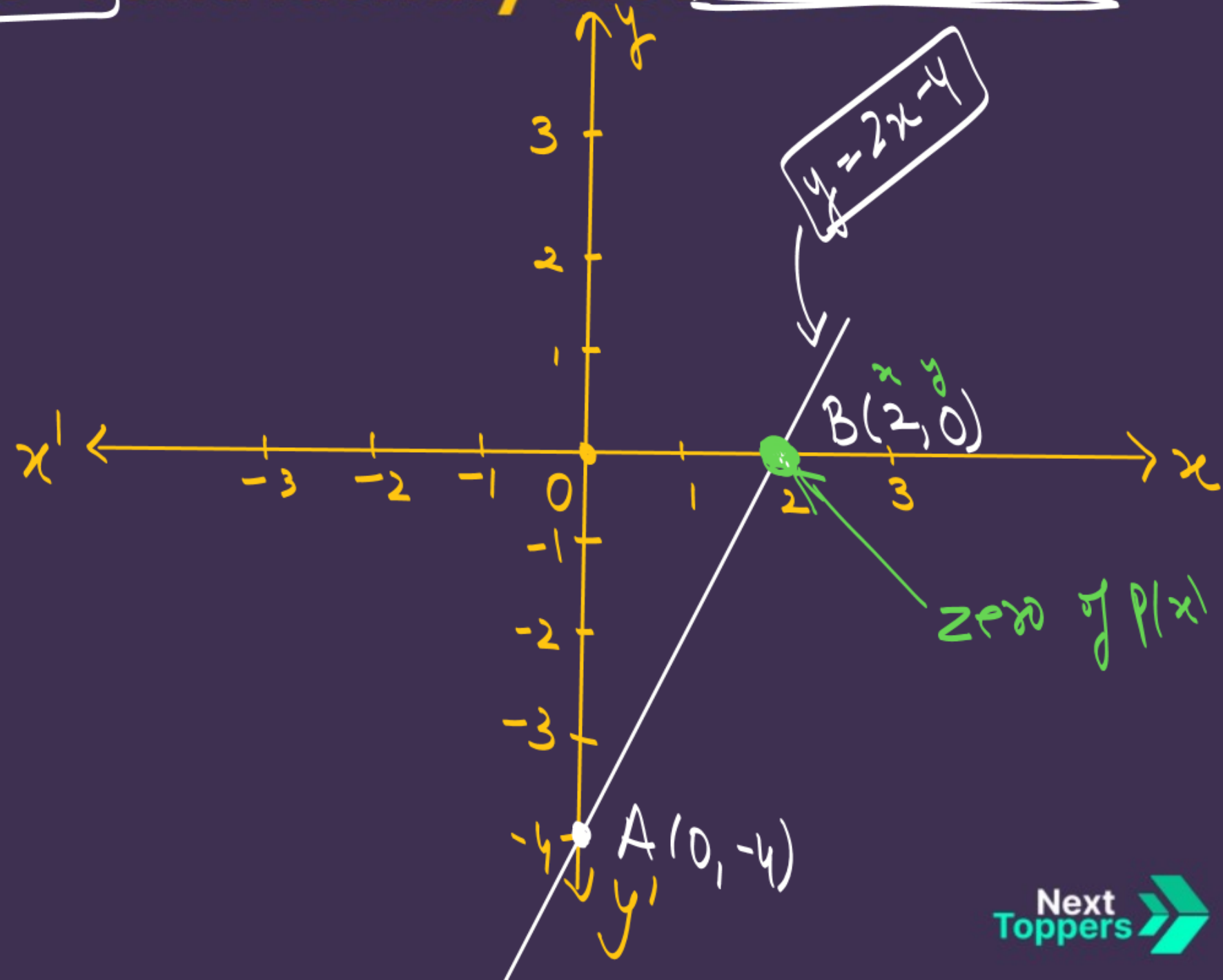
$$P(x) = 2x - 4$$

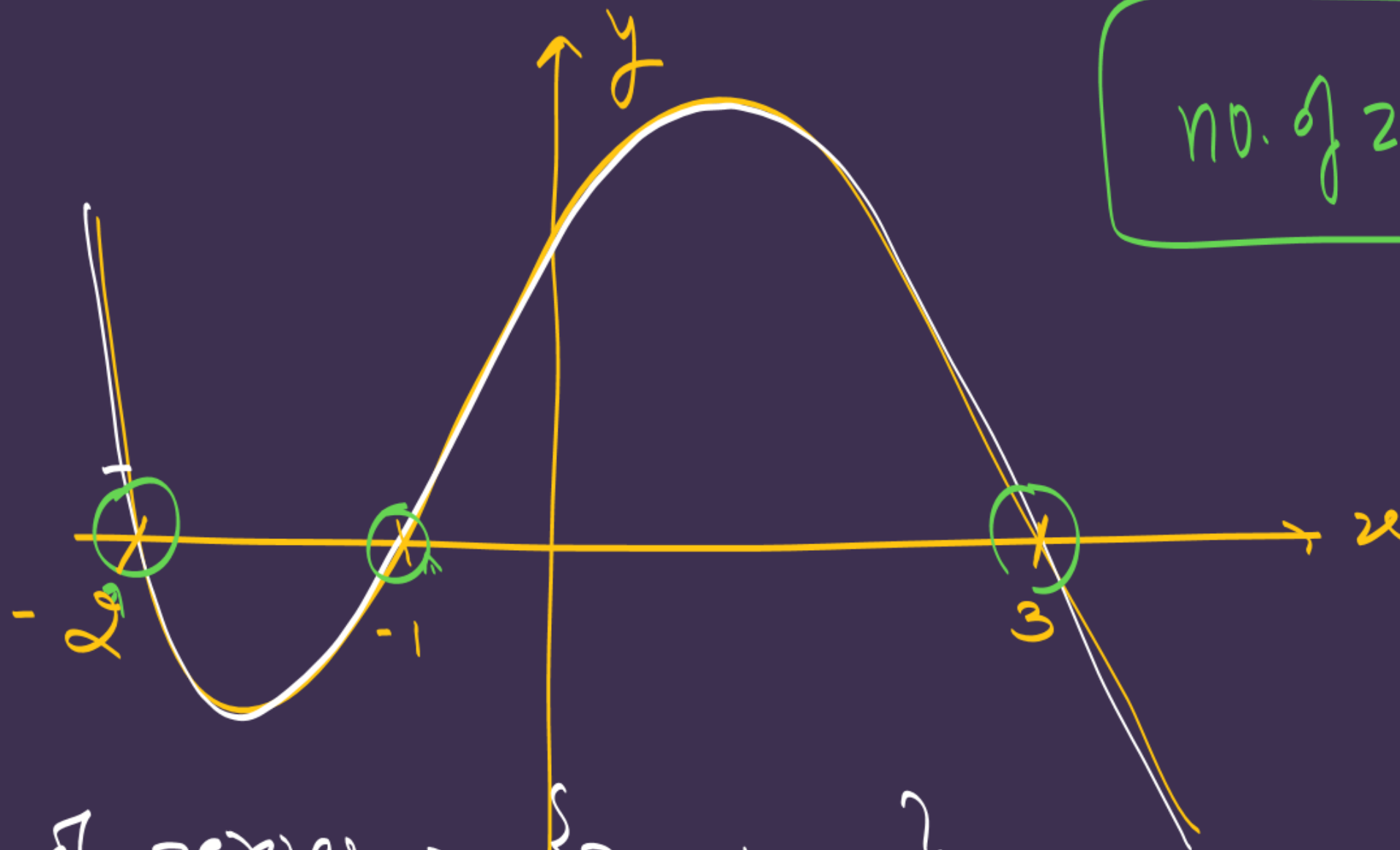
$$y = 2x - 4$$

|   |    |   |
|---|----|---|
| x | 0  | 2 |
| y | -4 | 0 |

$$\left. \begin{array}{l} A(0, -4) \\ B(2, 0) \end{array} \right\}$$

$$\begin{array}{l|l} y = 2(0) - 4 & 0 = 2x - 4 \\ \hline y = -4 & 24 = 2x \\ & x = 2 \end{array}$$

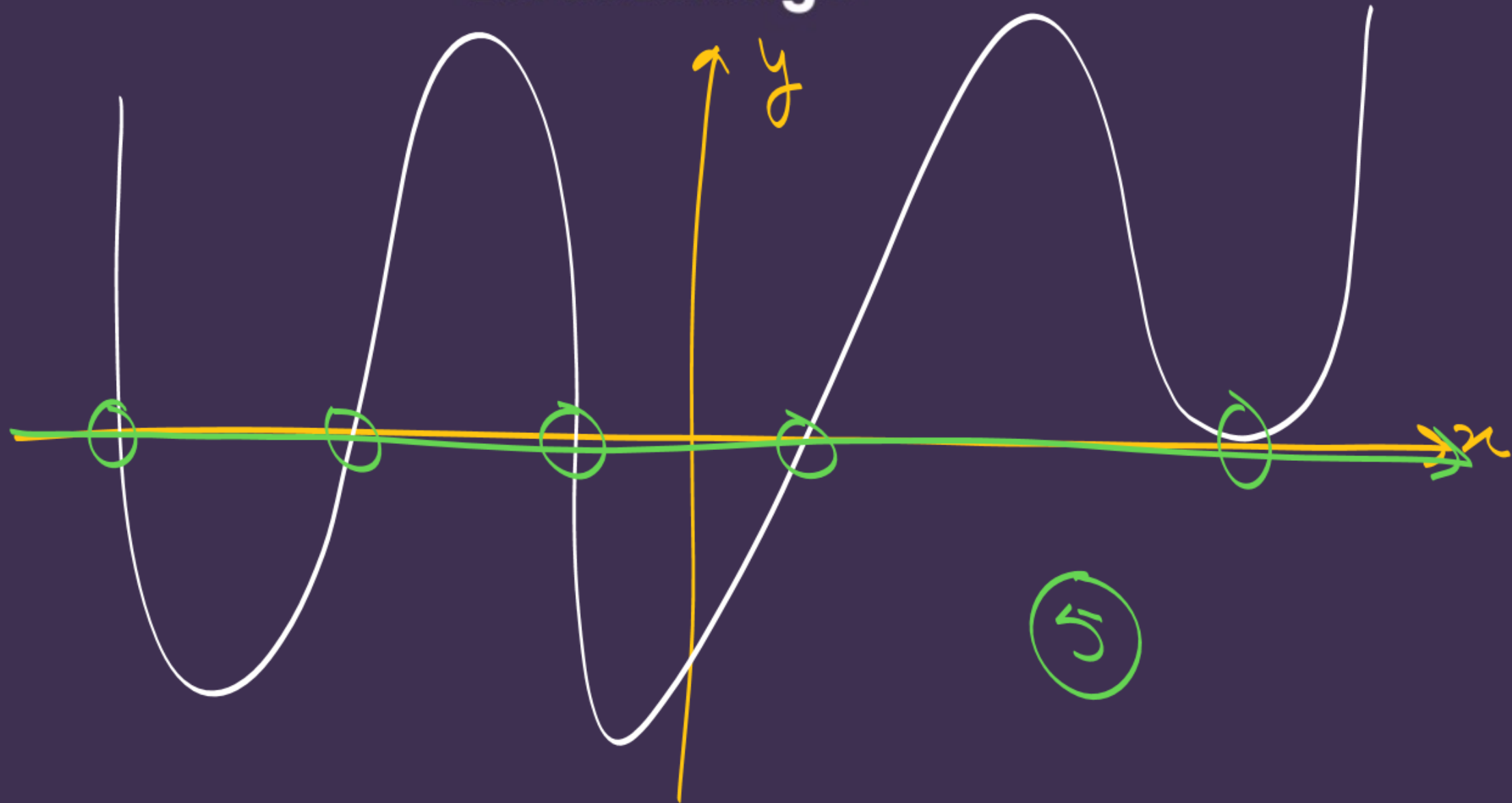




Value of zeroes  $\Rightarrow \underline{\underline{\{3, -1, -2\}}}$

# # No. of Zeroes of Polynomials by Graph:

Polynomial ka graph jitni baar x-axis ko touch/ intersect karega utne no. of zeroes honge





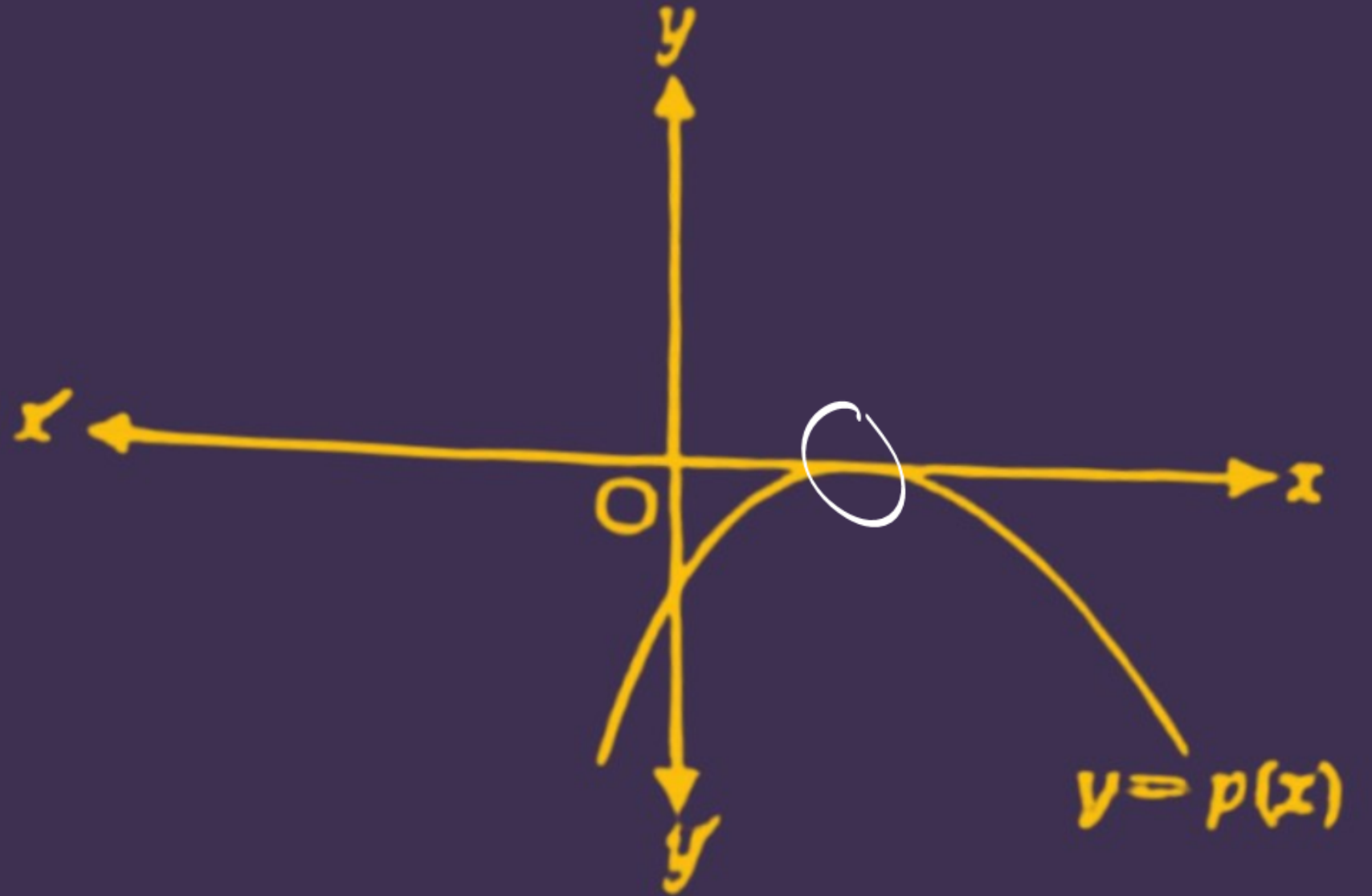
**#LP: The graph of  $y = p(x)$  is given, for a polynomial  $p(x)$ . The number of zeroes of  $p(x)$  from the graph is: [CBSE 2023]**

**(A) 3**

**(B) 1**

**(C) 2**

**(D) 0**





# 1 POLYNOMIAL & DEGREE

- **Definition:** A polynomial is an algebraic expression of the form

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \quad (n \geq 0)$$

- **Degree:** Highest power of  $x$  in the polynomial.
- **Key Point:** Degree determines the type of polynomial.



# 3 ZEROES OF POLYNOMIAL

- **Definition:** A value of  $x$  for which  $p(x) = 0$  is called a zero (or root) of the polynomial  $p(x)$ .
- **Key Point:** Zeroes of a polynomial are the  $x$ -intercepts of its graph.
- **Method:** We find zeroes by factorizing the polynomial and equating each factor to zero.

If  $p(x) = a(x - r_1)(x - r_2) \dots (x - r_n)$ , then zeroes are  $r_1, r_2, \dots, r_n$



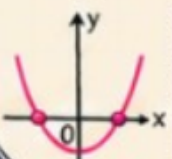
# 5 DISCRIMINANT & NATURE OF ROOTS (ZEROES)

For a quadratic polynomial  $ax^2 + bx + c$  ( $a \neq 0$ )

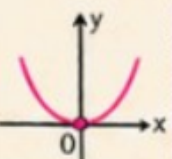
**Discriminant:**

$$D = b^2 - 4ac$$

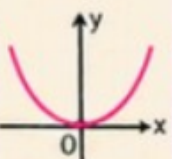
$D > 0$



$D = 0$



$D < 0$



**Nature of Roots (Zeroes):**

- If  $D > 0 \Rightarrow$  Two distinct real zeroes (graph cuts  $x$ -axis at two points)
- If  $D = 0 \Rightarrow$  Two equal real zeroes (graph touches  $x$ -axis at one point)
- If  $D < 0 \Rightarrow$  No real zeroes (graph does not cut or touch  $x$ -axis)

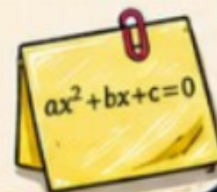
# 6 QUADRATIC EQUATION

**Definition:** An equation of the form  $ax^2 + bx + c = 0$ , where  $a, b, c$  are real numbers and  $a \neq 0$ , is called a quadratic equation in one variable.

**Solutions / Roots:** The values of  $x$  that satisfy  $ax^2 + bx + c = 0$  are called the roots (or solutions) of the equation.

**Formula:**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad a \neq 0$$

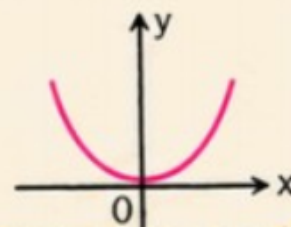


# 2 TYPES OF POLYNOMIALS

- **Constant Polynomial:**  $f(x) = k$  (degree 0)
- **Linear Polynomial:**  $f(x) = ax + b, a \neq 0$  (degree 1)
- **Quadratic Polynomial:**  $f(x) = ax^2 + bx + c, a \neq 0$  (degree 2)

**Examples:**

$$3, 7x - 4, 3x + 5, 4x^2 + 7x + 3$$



# 4 RELATION BETWEEN ZEROES & COEFFICIENTS

- **Linear Polynomial:**

$$ax + b \Rightarrow \text{Zero} = -\frac{b}{a}, \quad (a \neq 0)$$

- **Quadratic Polynomial:**

$$ax^2 + bx + c, \quad (a \neq 0)$$

Sum of zeroes =  $-\frac{b}{a}$

Product of zeroes =  $\frac{c}{a}$

# 7 MAKING OF QUADRATIC EQUATION

A quadratic equation can be formed if we know:

- **Two zeroes (roots):** If zeroes are  $\alpha$  and  $\beta$ ,  
Equation is  $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
- **One zero  $\alpha$  and coefficient of  $x^2$  is  $a$ :**  
Equation is  $ax^2 - a\alpha x + c = 0$   
(where  $c$  is any real number)
- **Sum ( $S = \alpha + \beta$ ) and product ( $P = \alpha\beta$ ) of zeroes:**  
Equation is  $x^2 - Sx + P = 0$



**THANK YOU COODIESS !!**

